



«Approved»
by the Dean

Syzydykova Z.A.
«__» of _____ 2025

Syllabus Academic Year 2024-2025

1. General Information	
Course title	Neural Networks
Degree cycle (level)/major	Master's degree, Computer science and engineering(CSE)
Year, term	1
Number of credits	5
Language of delivery:	English, Russian
Prerequisites	There are no specific prerequisites or experience required for the course .However, it will be only beneficial if students have a solid understanding if programming, machine leaning
Postrequisites	More advanced courses on Artificial intelligence and machine learning as well as projects (including master's dissertation) / case studies related to students' major.
Lecturer(s)	Niyazova Rozamgul, The Candidate of Technical Sciences, Asistent Professor, Rozamgul.Niyazova@astanait.edu.kz, Astana IT University, Expo, C1 block, 2nd floor, office C1.1.332).
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	This course provides a comprehensive introduction to neural networks, covering foundational concepts, architectures, and practical applications. Topics include the perceptron model, activation functions, backpropagation, optimization, and regularization techniques. Advanced topics such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative adversarial networks (GANs) are explored alongside their applications in image processing, natural language processing, and time-series analysis. Students will gain hands-on experience through coding assignments and independent projects, preparing them for real-world challenges in applied data analytics and computational
2. Course Goal(s)	The course aims to provide students with a strong foundation in neural network theory and practice, enabling them to design, implement, and evaluate neural network models for diverse applications in data analytics and computational sciences. Students will develop critical problem-solving skills and gain hands-on experience with modern tools and techniques, preparing them for advanced research or industry roles in artificial intelligence and machine learning.
3. Course Objectives:	Introduce the fundamental principles of neural networks and their applications. Explore key neural network architectures and their use cases. Develop skills in training and optimizing neural networks. Enable practical implementation using modern machine learning frameworks. Foster critical thinking through independent study and project work.

4. Skills & Competences	Technical Skills: Design and train neural networks using TensorFlow and PyTorch. Preprocess data and handle large datasets. Implement CNNs, RNNs, and GANs. Optimize models with advanced techniques. Evaluate and deploy neural network models efficiently. Problem-Solving Skills: Capability to apply neural networks to solve real-world problems in areas like data analytics, natural language processing, and image processing. Creativity in developing novel solutions using advanced neural network techniques. Research and Development: Ability to conduct independent research on emerging trends in neural networks. Skill in implementing and evaluating state-of-the-art architectures. Analytical Skills: Select appropriate neural network architectures for specific tasks. Analyze and interpret model performance metrics. Identify and address issues like overfitting and bias. Optimize neural networks for improved accuracy and efficiency.
5. Course Learning Outcomes:	By the end of this course students will be able to: Demonstrate understanding of neural network fundamentals and architectures. Design, implement, and train neural network models using modern frameworks. Apply neural networks to solve real-world problems in data analytics and computational sciences. Analyze and optimize model performance using appropriate techniques. Conduct independent research on advanced neural network topics. Address ethical challenges and biases in neural network applications.
6. Methods of Assessment	The expected learning outcomes for the course will be assessed through graded activities and ungraded activities. The graded activities include exams, homework, quizzes, and midterms. The ungraded activities will be used to monitor your progress. A variety of these ungraded assessment techniques may be employed, including problems to be completed during class, direct questioning of students, answering questions in class and discussions during office hours.
7. Reading List	Assigned Class lectures and discussions will proceed with supplemental and advanced topics reading materials and presentations should be read prior to class., which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1) Deep Learning by Ian Goodfellow, Yoshua Bengio, and Aaron Courville ISBN: 978-0262035613 2) Neural Networks and Deep Learning: A Textbook by Charu C. Aggarwal ISBN: 978-3319944623 3) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron ISBN: 978-1492032649

	4) Deep Learning with Python by François Chollet ISBN: 978-1617294433 5) Pattern Recognition and Machine Learning by Christopher M. Bishop ISBN: 978-0387310732 6) Introduction to Neural Networks for Java and Python by Jeff Heaton ISBN: 978-1604390260
1. Resources	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
2. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student’s misconduct, the student can be dispensed from the classes. Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor.

Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Final exam* / Final project:

Test

An individual (team) project (report) and presentation (slides). At the completion of this course each student /each team has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline.

Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor.

Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround.

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

1. Submitting work that is not your own papers, assignments, or exams;
2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;
3. Submitting or using falsified data;
4. Submitting the same work for credit in two courses without prior consent of both instructors.

Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university.

Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using

	Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Introduction to neural networks: Basic concepts, biological motivation, perceptron architecture, binary classification, visualization of activation functions.	2	3		1	7
2	Activation Functions and Learning: Sigmoid, ReLU, Tanh, Softmax; gradient descent, backpropagation, multilayer perceptron (MLP), hyperparameter tuning.	2	3		1	7
3	Optimization and Regularization: SGD, Adam, RMSprop; Dropout, L1/L2 regularization; Grid Search, Random Search; CNN for image classification, Transfer Learning.	2	3		1	7
4	Convergence Analysis and Overfitting Problem: Factors affecting convergence, overfitting, techniques for mitigation (early stopping, regularization, data augmentation).	2	3		1	7
5	Recurrent Neural Networks and Autoencoders: RNN, LSTM, GRU; autoencoders for image compression, data generation with GANs.	2	3		1	7
6	Deep Learning and Generative Models: Transfer Learning, GANs, time series analysis, text classification.	2	3		1	7
7	Neural Network Training Techniques and Best Practices: Batch normalization, dropout, data augmentation, transfer learning strategies, fine-tuning deep models.	2	3		1	7

8	Interpretability and Explainability of Neural Networks: Interpretation methods (LIME, SHAP), CNN filter visualization, internal representations analysis.	2	3		1	7
9	Normalization of Input Data: Vanishing/exploding gradients, initialization for deep networks, numerical gradient approximation.	2	3		1	7
10	Applications of Neural Networks in Medicine: Neural networks for image and video analysis.	2	3		1	7
	Total hours: 130	20	30		10	70

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Exploration of Neural Network Architectures Research modern neural network architectures like ResNet, DenseNet, and EfficientNet. Write a report comparing their structures and applications.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
2	Comparative Study of Optimization Algorithms Compare optimization algorithms (e.g., SGD, Adam, RMSprop) on different datasets. Analyze the convergence rates and final model accuracy.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
3	Designing Custom Activation Functions Create and test a new activation function. Analyze its performance against standard functions like ReLU and Tanh.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
4	Neural Networks in Natural Language Processing (NLP) Study the role of word embeddings like Word2Vec and GloVe in NLP. Implement a text classification task using embeddings.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
5	Unsupervised Learning with Neural Networks Explore autoencoders for clustering tasks. Compare results with traditional	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version

	clustering algorithms (e.g., K-means).			
6	Applications of GANs Investigate the use of GANs in generating synthetic images or data. Study challenges such as mode collapse and how to address them.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
7	Physics-Informed Neural Networks (PINNs) Study how PINNs solve differential equations in physics and engineering. Propose a novel application area for PINNs.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
8	Explainability in Neural Networks Research explainability techniques like SHAP and LIME. Apply these techniques to a trained model and interpret the results.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
9	Integration of Neural Networks with Reinforcement Learning Study the use of neural networks as function approximators in reinforcement learning. Implement a simple reinforcement learning task using neural networks.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
10	Ethics and Bias in Neural Networks Research bias in neural network datasets and algorithms. Propose techniques to mitigate bias in model training and evaluation.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignments**: assignment 1, assignment 2, assignment 3, Mid Term	60 20 20 20 40	100

2 nd attestation	Assignments**: assignment 4, assignment 5, assignment 6, End Term	60 20 20 20 40	100
Final exam*	Written Exam/Quiz/IP/TP		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

*** The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.*

** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment

80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Rozamgul Niyazova		14/03/2025	

Director

Sergaziyev M.Zh.